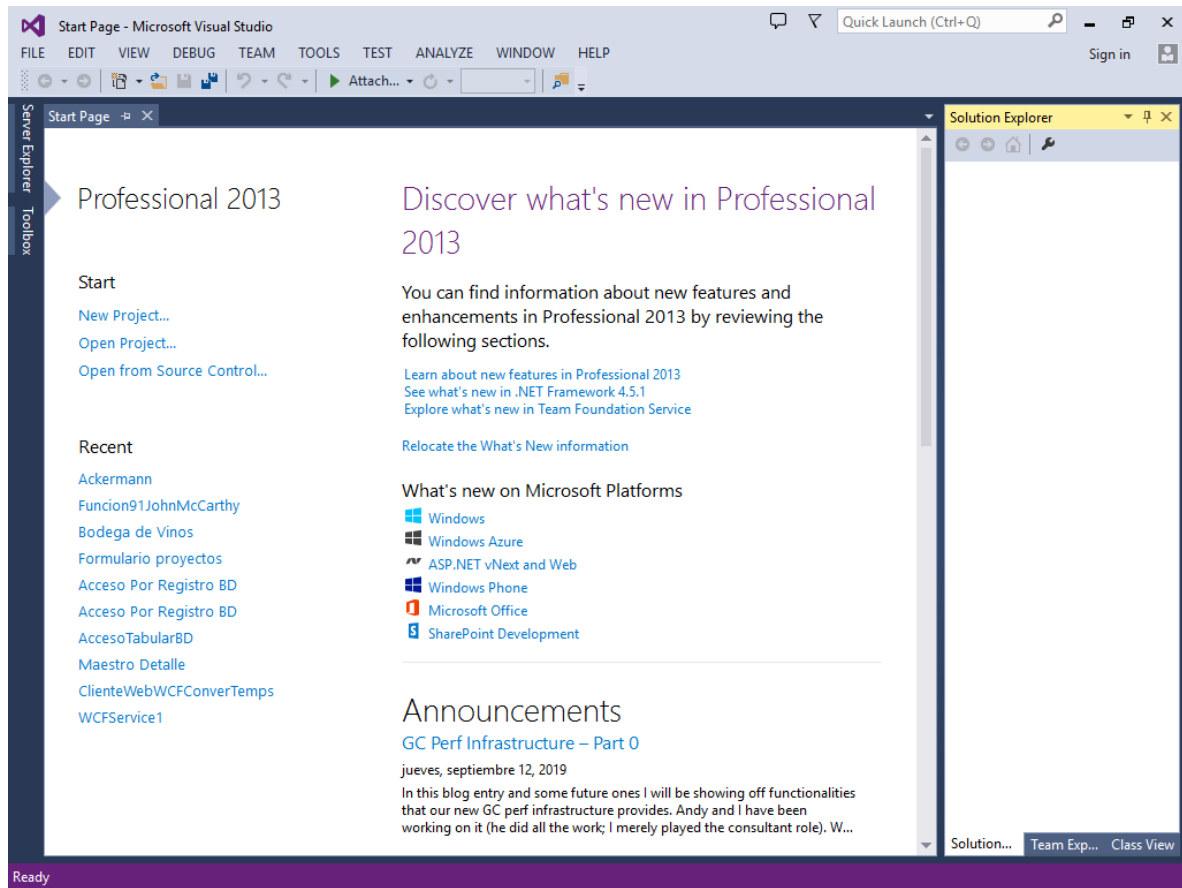
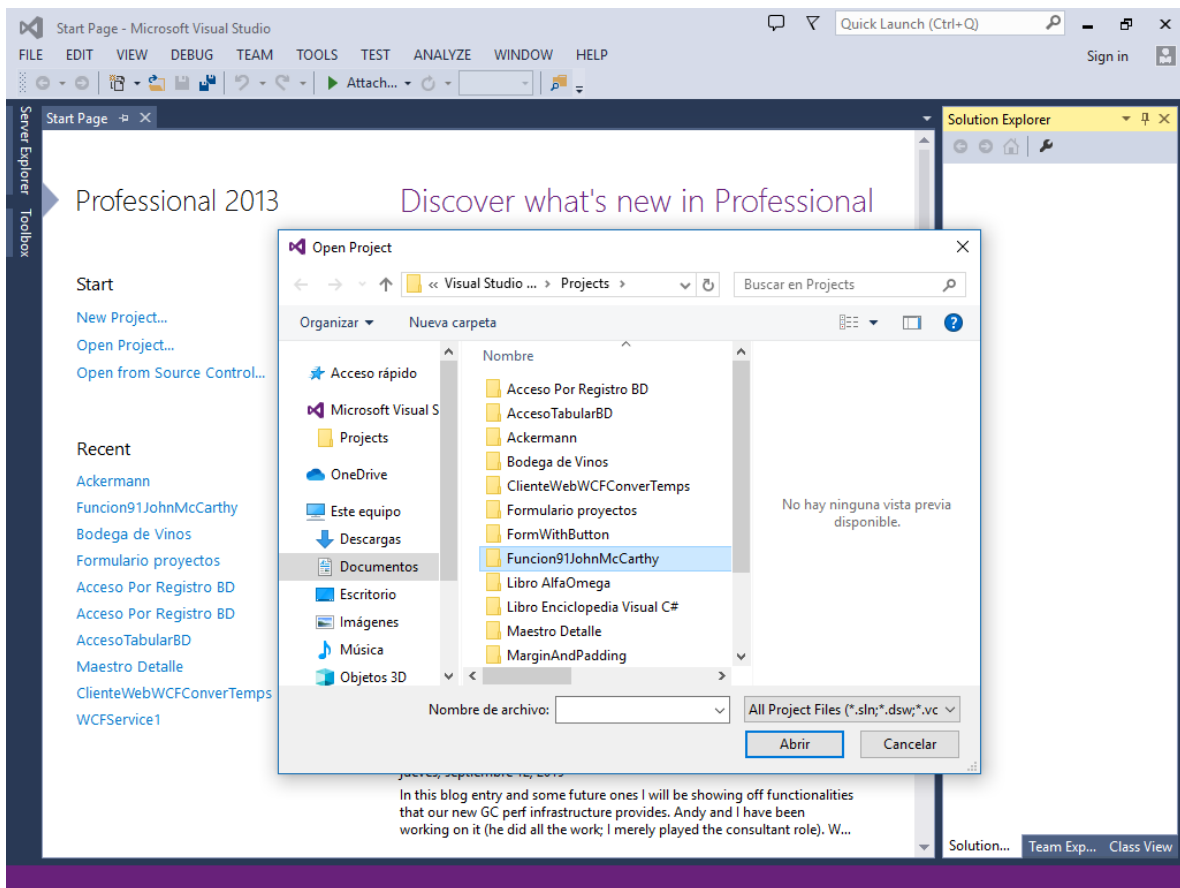
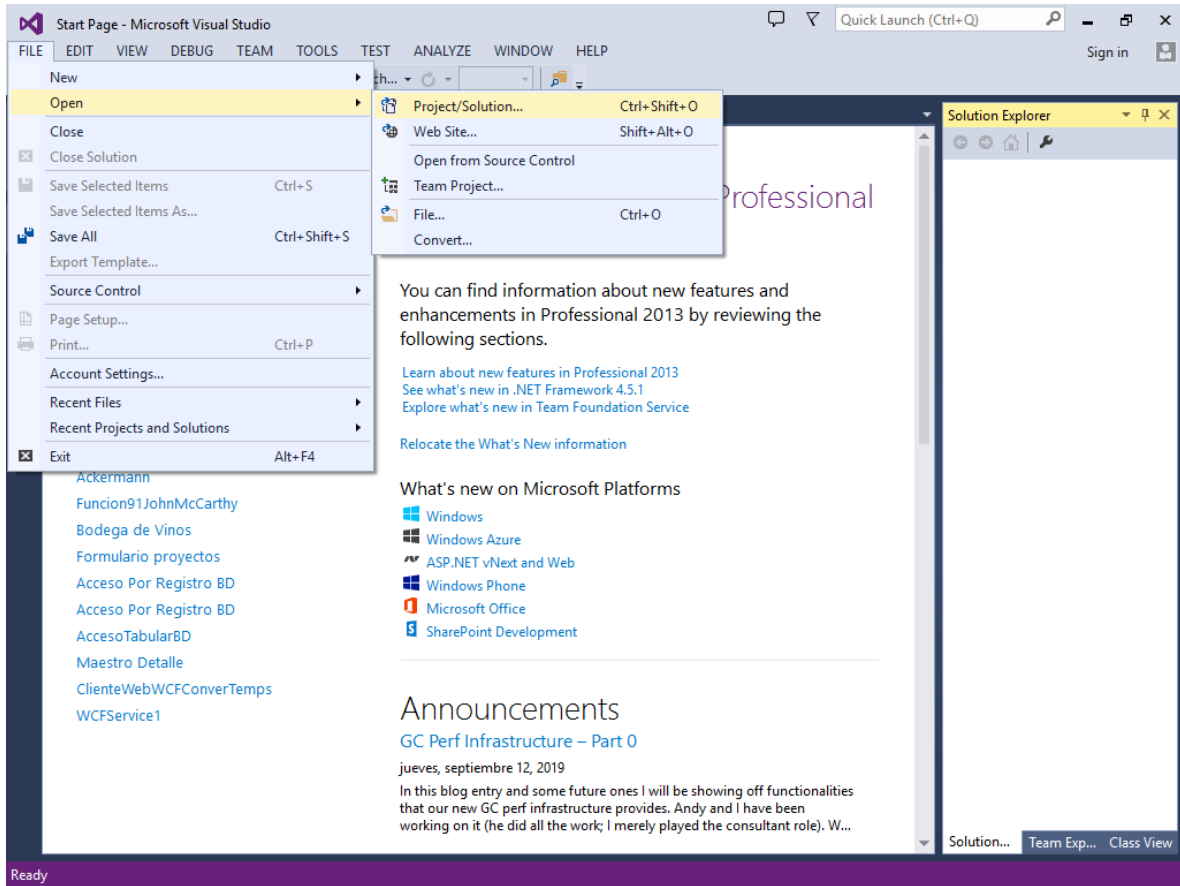
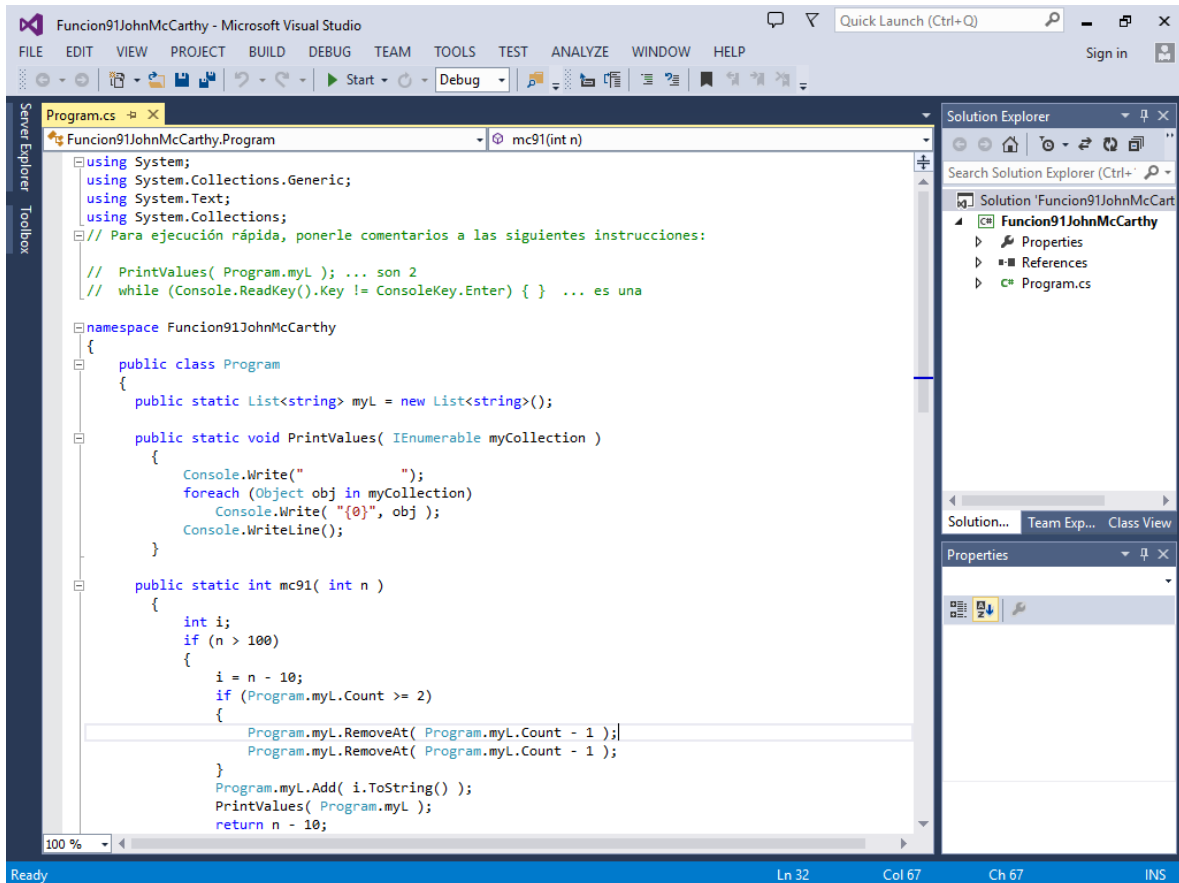
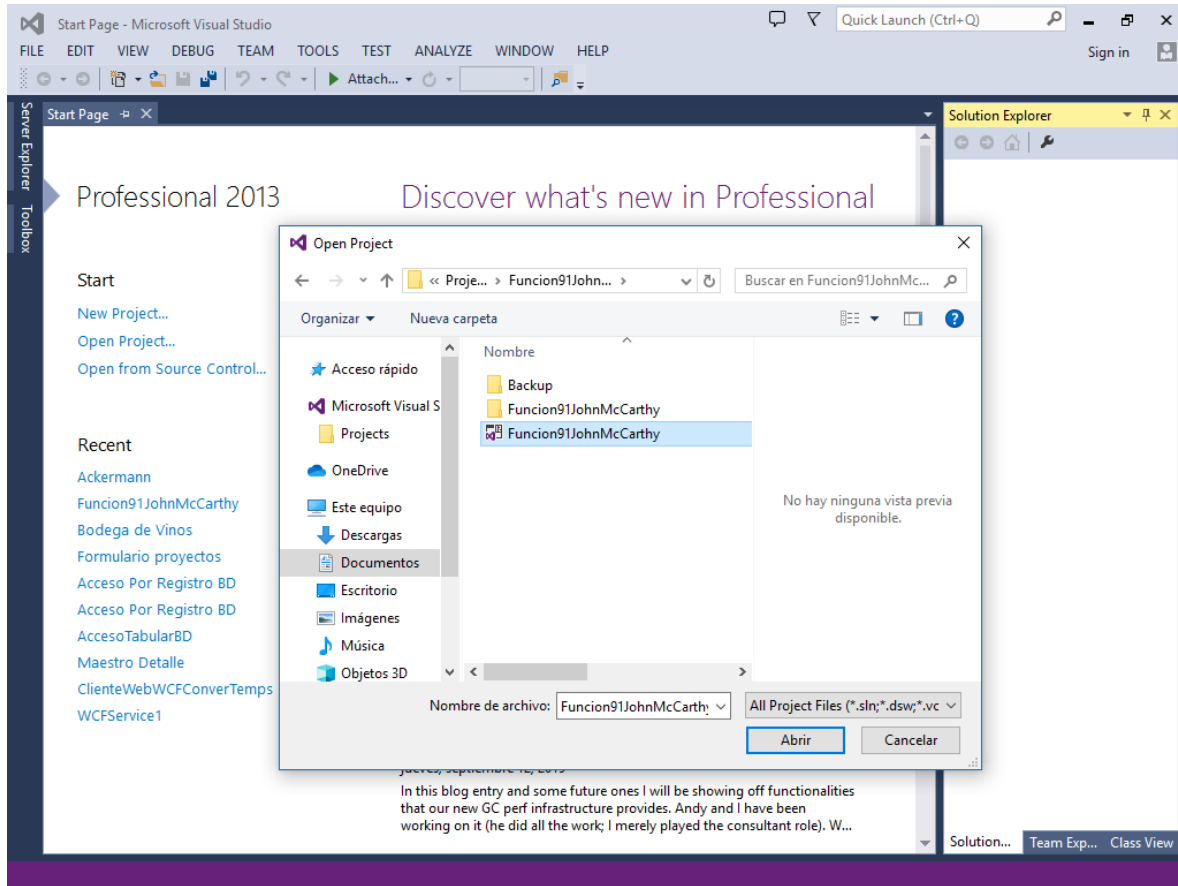


Programación de la función 91 de John McCarthy en Visual Studio







```

using System;
using System.Collections.Generic;
using System.Text;
using System.Collections;
// Para ejecución rápida, ponerle comentarios a las siguientes instrucciones:

// PrintValues( Program.myL ); ... son 2
// while (Console.ReadKey().Key != ConsoleKey.Enter) { } ... es una

namespace Funcion91JohnMcCarthy
{
    public class Program
    {
        public static List<string> myL = new List<string>();

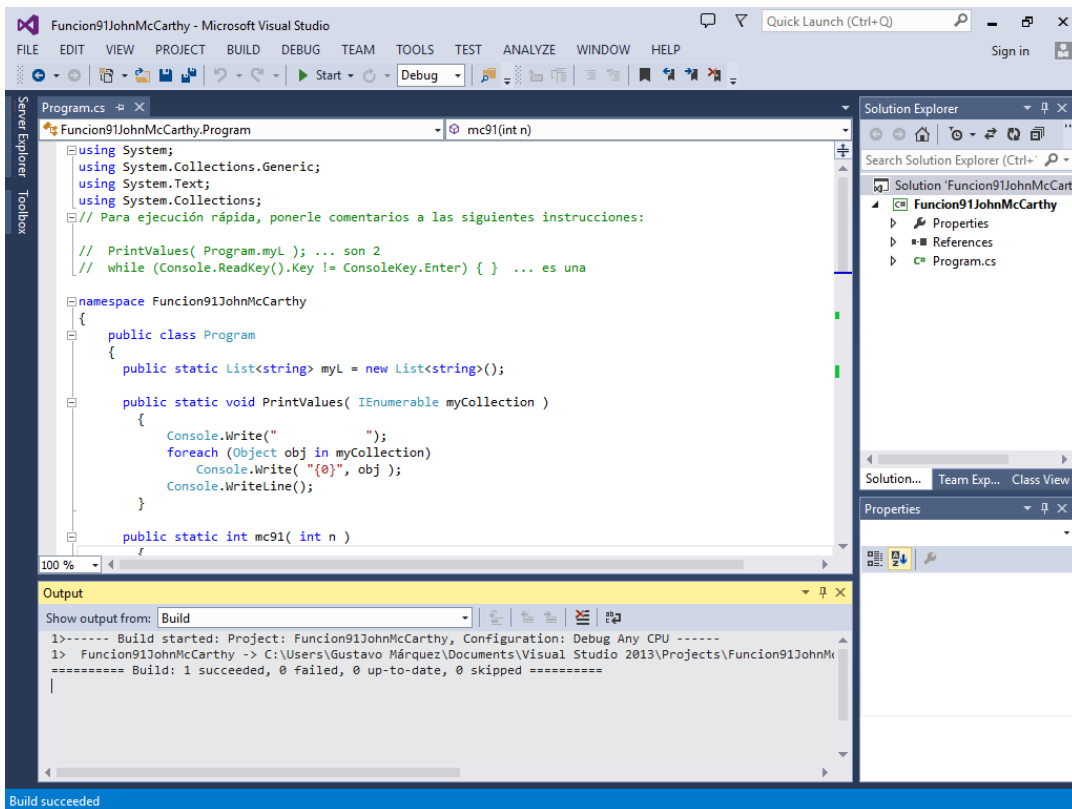
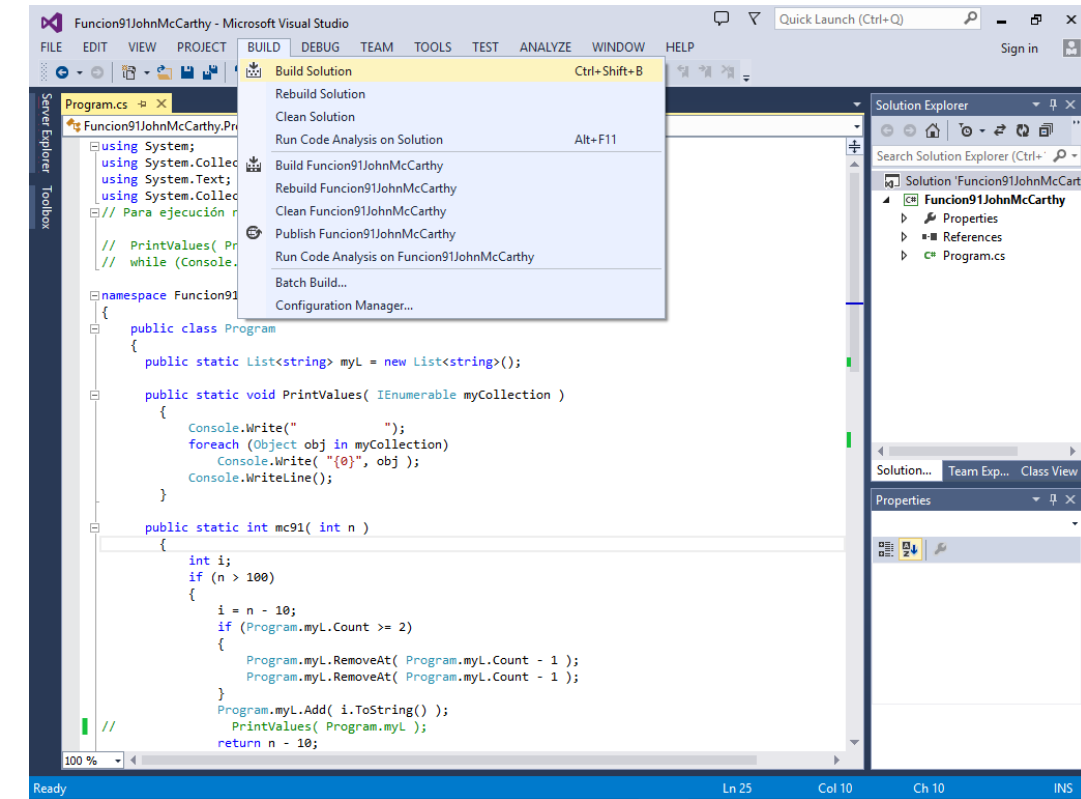
        public static void PrintValues( IEnumerable myCollection )
        {
            Console.Write(" ");
            foreach (Object obj in myCollection)
                Console.Write( "{0}", obj );
            Console.WriteLine();
        }

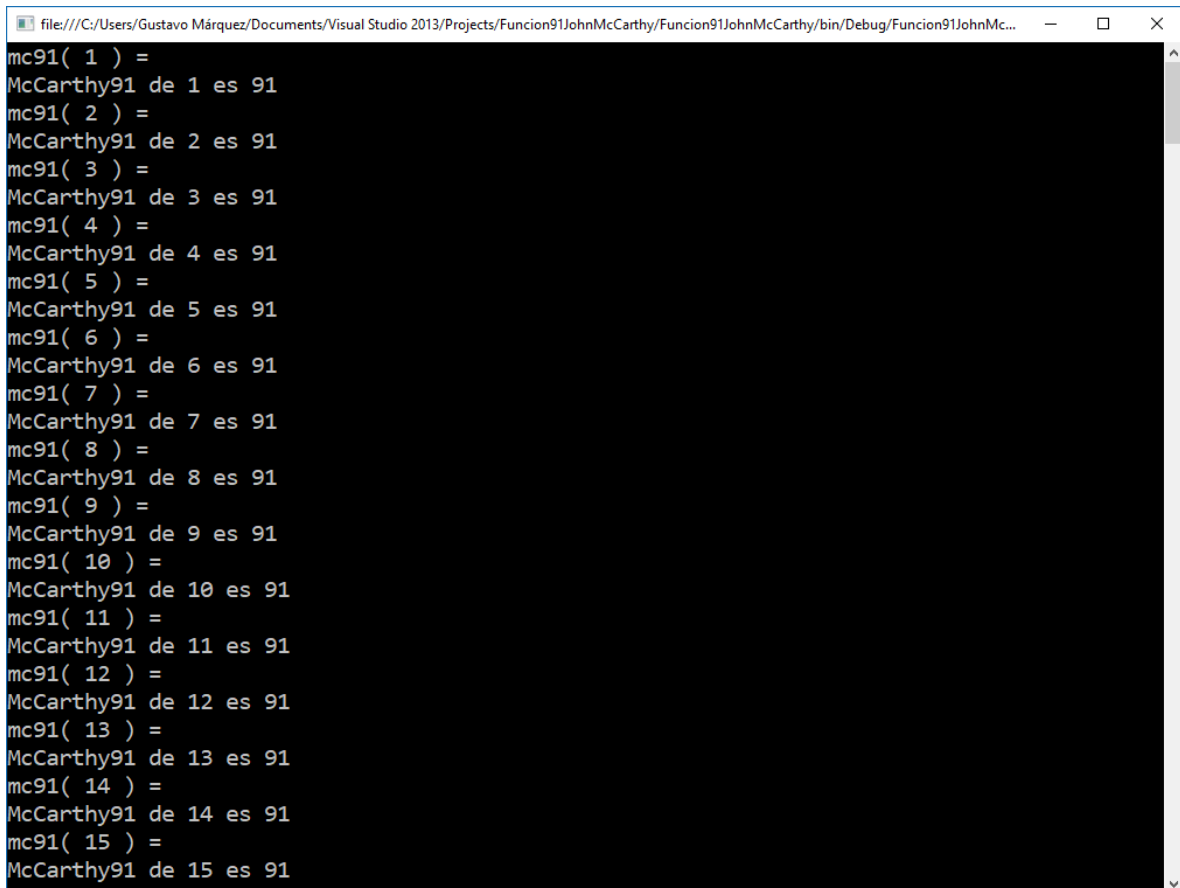
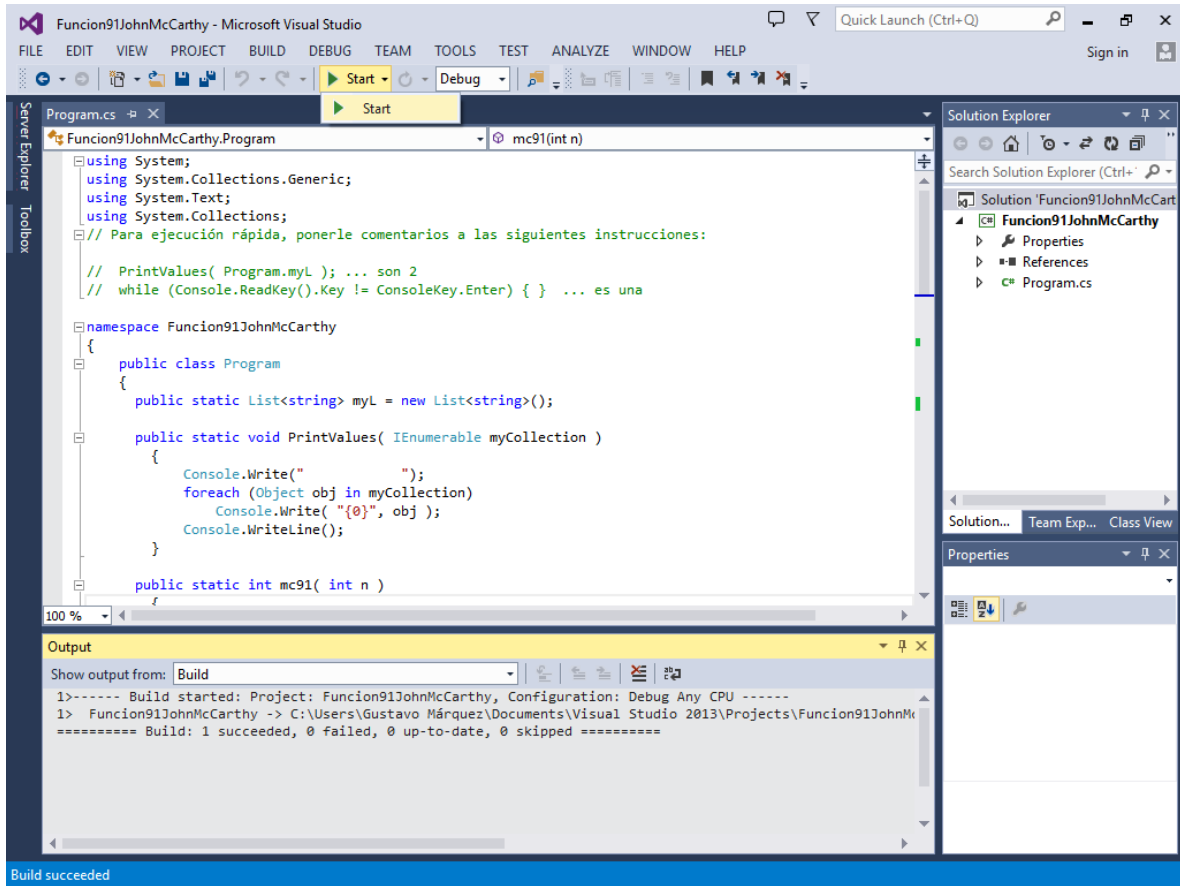
        public static int mc91( int n )
        {
            int i;
            if (n > 100)
            {
                i = n - 10;
                if (Program.myL.Count >= 2)
                {
                    Program.myL.RemoveAt( Program.myL.Count - 1 );
                    Program.myL.RemoveAt( Program.myL.Count - 1 );
                }
                Program.myL.Add( i.ToString() );
                PrintValues( Program.myL );
                return n - 10;
            }
            else
            {
                i = n + 11;
                if (Program.myL.Count >= 2)
                {
                    Program.myL.RemoveAt( Program.myL.Count - 1 );
                    Program.myL.RemoveAt( Program.myL.Count - 1 );
                }
                Program.myL.Add( "mc91 " );
                Program.myL.Add( "mc91 " );
                Program.myL.Add( i.ToString() );
                PrintValues( Program.myL );
                while (Console.ReadKey().Key != ConsoleKey.Enter) { }
                return mc91( mc91( n + 11 ) );
            }
        }

        static void Main( string[] args )
        {
            for (int i = 1; i <= 120; i++)
            {
                Console.WriteLine( "mc91( " + i + " ) =" );
                Console.WriteLine( "McCarthy91 de " + i + " es " + Program.mc91(i) );
                myL.Clear();
            }
        }
    }
}

```

```
Console.ReadKey();
```

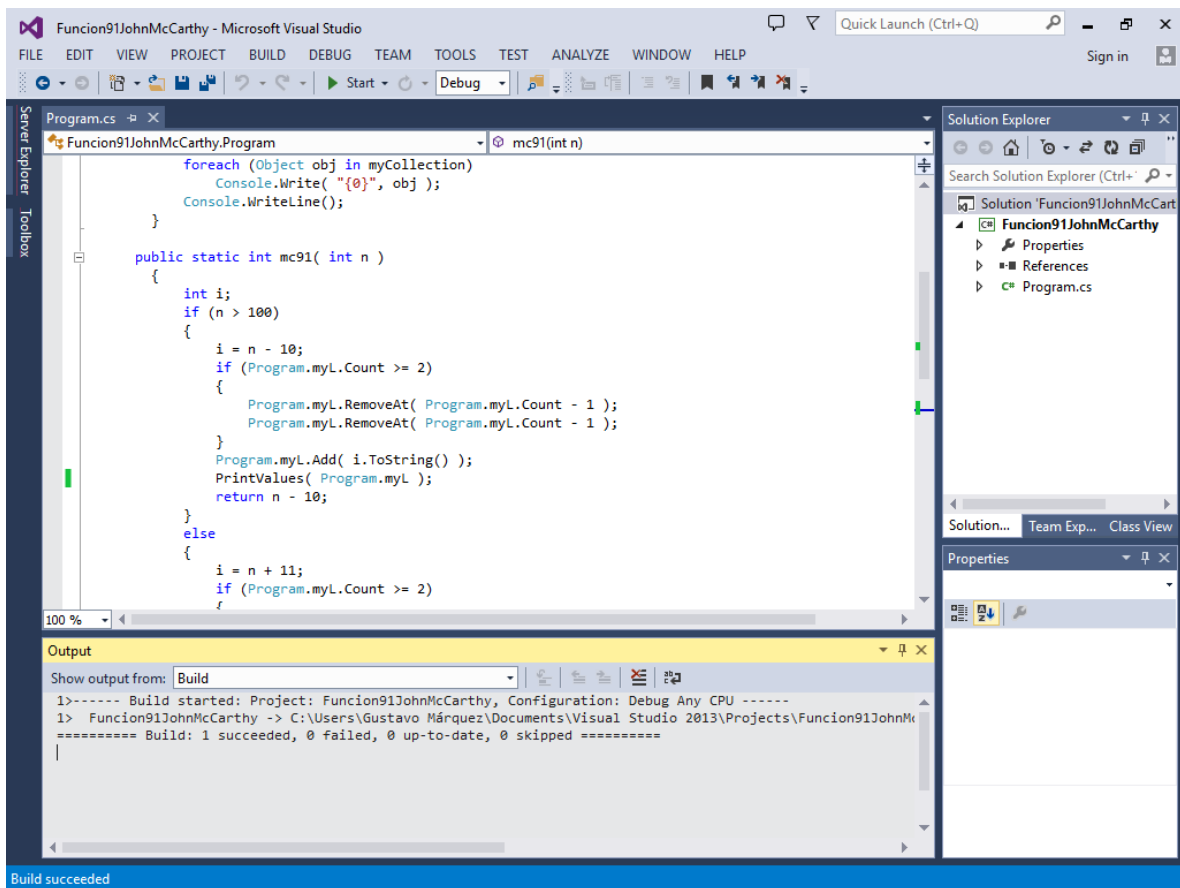
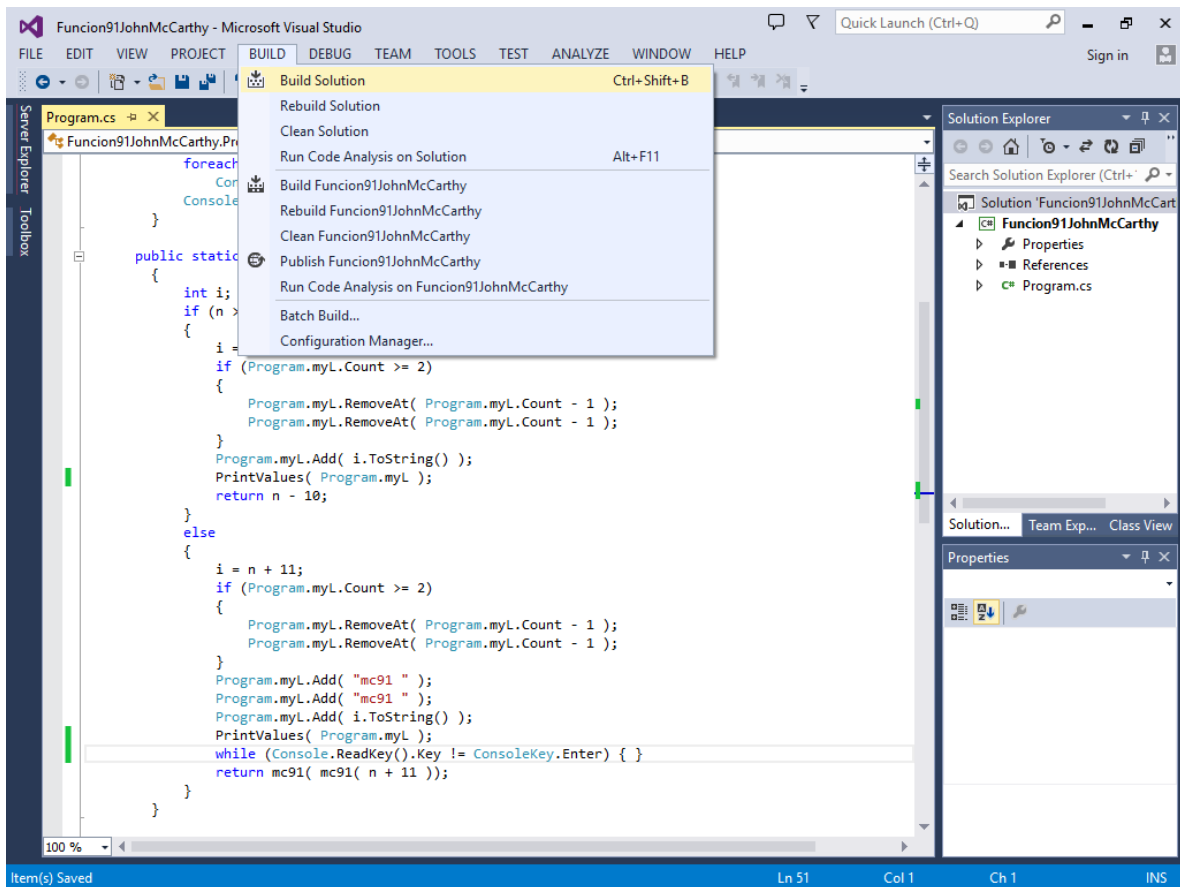


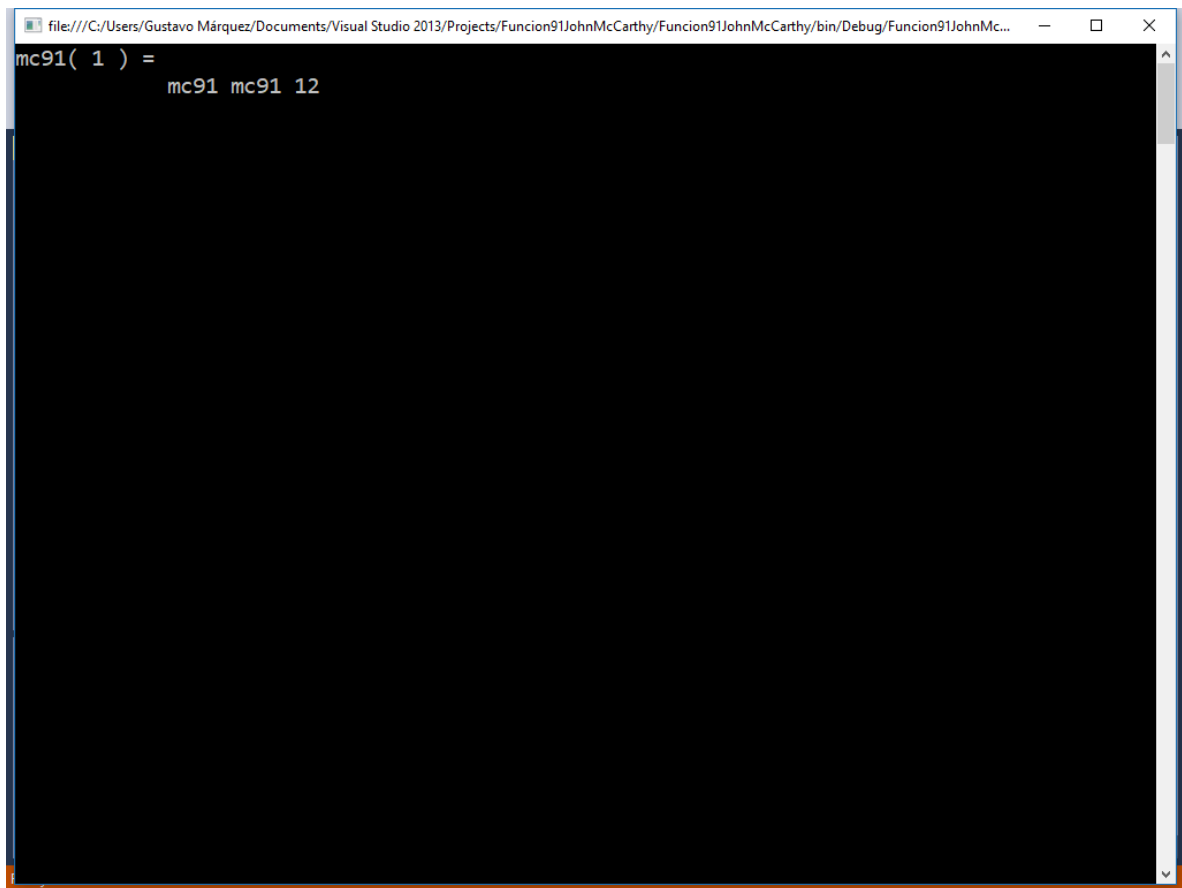
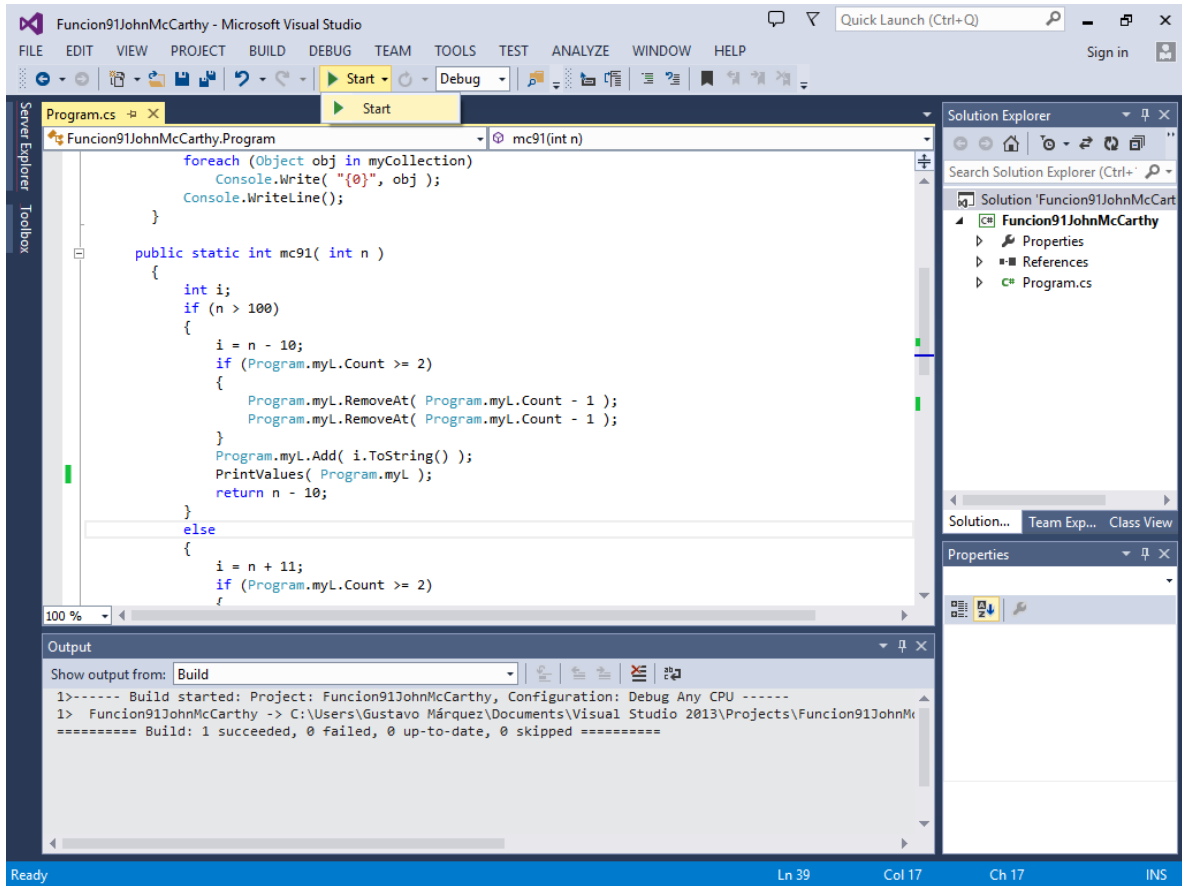


```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
McCarthy91 de 93 es 91
mc91( 94 ) =
McCarthy91 de 94 es 91
mc91( 95 ) =
McCarthy91 de 95 es 91
mc91( 96 ) =
McCarthy91 de 96 es 91
mc91( 97 ) =
McCarthy91 de 97 es 91
mc91( 98 ) =
McCarthy91 de 98 es 91
mc91( 99 ) =
McCarthy91 de 99 es 91
mc91( 100 ) =
McCarthy91 de 100 es 91
mc91( 101 ) =
McCarthy91 de 101 es 91
mc91( 102 ) =
McCarthy91 de 102 es 92
mc91( 103 ) =
McCarthy91 de 103 es 93
mc91( 104 ) =
McCarthy91 de 104 es 94
mc91( 105 ) =
McCarthy91 de 105 es 95
mc91( 106 ) =
McCarthy91 de 106 es 96
mc91( 107 ) =
McCarthy91 de 107 es 97
mc91( 108 ) =
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91( 107 ) =
McCarthy91 de 107 es 97
mc91( 108 ) =
McCarthy91 de 108 es 98
mc91( 109 ) =
McCarthy91 de 109 es 99
mc91( 110 ) =
McCarthy91 de 110 es 100
mc91( 111 ) =
McCarthy91 de 111 es 101
mc91( 112 ) =
McCarthy91 de 112 es 102
mc91( 113 ) =
McCarthy91 de 113 es 103
mc91( 114 ) =
McCarthy91 de 114 es 104
mc91( 115 ) =
McCarthy91 de 115 es 105
mc91( 116 ) =
McCarthy91 de 116 es 106
mc91( 117 ) =
McCarthy91 de 117 es 107
mc91( 118 ) =
McCarthy91 de 118 es 108
mc91( 119 ) =
McCarthy91 de 119 es 109
mc91( 120 ) =
McCarthy91 de 120 es 110
```

Eliminando los comentarios y ejecutando el programa:






```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91 mc91 mc91 mc91 mc91 91
mc91 mc91 mc91 mc91 mc91 mc91 102
mc91 mc91 mc91 mc91 mc91 92
mc91 mc91 mc91 mc91 mc91 mc91 103
mc91 mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 mc91 mc91 105
mc91 mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 mc91 mc91 106
mc91 mc91 mc91 mc91 mc91 96
mc91 mc91 mc91 mc91 mc91 mc91 107
mc91 mc91 mc91 mc91 mc91 97
mc91 mc91 mc91 mc91 mc91 mc91 108
mc91 mc91 mc91 mc91 mc91 98
mc91 mc91 mc91 mc91 mc91 mc91 109
mc91 mc91 mc91 mc91 mc91 99
mc91 mc91 mc91 mc91 mc91 mc91 110
mc91 mc91 mc91 mc91 mc91 100
mc91 mc91 mc91 mc91 mc91 mc91 111
mc91 mc91 mc91 mc91 mc91 101
mc91 mc91 mc91 mc91 mc91 91
mc91 mc91 mc91 mc91 mc91 102
mc91 mc91 mc91 mc91 mc91 92
mc91 mc91 mc91 mc91 mc91 103
mc91 mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 mc91 105
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91 mc91 mc91 mc91 92
mc91 mc91 mc91 mc91 mc91 103
mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 mc91 105
mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 mc91 106
mc91 mc91 mc91 mc91 96
mc91 mc91 mc91 mc91 mc91 107
mc91 mc91 mc91 mc91 97
mc91 mc91 mc91 mc91 mc91 108
mc91 mc91 mc91 mc91 98
mc91 mc91 mc91 mc91 mc91 109
mc91 mc91 mc91 mc91 99
mc91 mc91 mc91 mc91 mc91 110
mc91 mc91 mc91 mc91 100
mc91 mc91 mc91 mc91 mc91 111
mc91 mc91 mc91 mc91 101
mc91 mc91 mc91 mc91 91
mc91 mc91 mc91 mc91 102
mc91 mc91 mc91 mc91 92
mc91 mc91 mc91 mc91 103
mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 105
mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 106
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...  
mc91 mc91 mc91 94  
mc91 mc91 mc91 mc91 105  
mc91 mc91 mc91 mc91 95  
mc91 mc91 mc91 mc91 106  
mc91 mc91 mc91 mc91 96  
mc91 mc91 mc91 mc91 107  
mc91 mc91 mc91 mc91 97  
mc91 mc91 mc91 mc91 108  
mc91 mc91 mc91 mc91 98  
mc91 mc91 mc91 mc91 109  
mc91 mc91 mc91 mc91 99  
mc91 mc91 mc91 mc91 110  
mc91 mc91 mc91 mc91 100  
mc91 mc91 mc91 mc91 111  
mc91 mc91 mc91 mc91 101  
mc91 mc91 mc91 91  
mc91 mc91 mc91 mc91 102  
mc91 mc91 mc91 92  
mc91 mc91 mc91 mc91 103  
mc91 mc91 mc91 93  
mc91 mc91 mc91 mc91 104  
mc91 mc91 mc91 94  
mc91 mc91 mc91 mc91 105  
mc91 mc91 mc91 95  
mc91 mc91 mc91 mc91 106  
mc91 mc91 mc91 96  
mc91 mc91 mc91 mc91 107  
mc91 mc91 mc91 97  
mc91 mc91 mc91 mc91 108
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...  
mc91 mc91 mc91 96  
mc91 mc91 mc91 mc91 107  
mc91 mc91 mc91 mc91 97  
mc91 mc91 mc91 mc91 108  
mc91 mc91 mc91 mc91 98  
mc91 mc91 mc91 mc91 109  
mc91 mc91 mc91 mc91 99  
mc91 mc91 mc91 mc91 110  
mc91 mc91 mc91 mc91 100  
mc91 mc91 mc91 mc91 111  
mc91 mc91 mc91 101  
mc91 mc91 mc91 91  
mc91 mc91 mc91 mc91 102  
mc91 mc91 mc91 92  
mc91 mc91 mc91 mc91 103  
mc91 mc91 mc91 93  
mc91 mc91 mc91 mc91 104  
mc91 mc91 mc91 94  
mc91 mc91 mc91 mc91 105  
mc91 mc91 mc91 95  
mc91 mc91 mc91 mc91 106  
mc91 mc91 mc91 96  
mc91 mc91 mc91 mc91 107  
mc91 mc91 mc91 97  
mc91 mc91 mc91 mc91 108  
mc91 mc91 mc91 98  
mc91 mc91 mc91 mc91 109  
mc91 mc91 mc91 99  
mc91 mc91 mc91 mc91 110
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91 mc91 mc91 110
mc91 mc91 100
mc91 mc91 mc91 111
mc91 mc91 101
mc91 91
mc91 mc91 102
mc91 92
mc91 mc91 103
mc91 93
mc91 mc91 104
mc91 94
mc91 mc91 105
mc91 95
mc91 mc91 106
mc91 96
mc91 mc91 107
mc91 97
mc91 mc91 108
mc91 98
mc91 mc91 109
mc91 99
mc91 mc91 110
mc91 100
mc91 mc91 111
mc91 101
91
McCarthy91 de 1 es 91
mc91( 2 ) =
    mc91 mc91 13
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
McCarthy91 de 1 es 91
mc91( 2 ) =
    mc91 mc91 13
    mc91 mc91 mc91 24
    mc91 mc91 mc91 mc91 35
    mc91 mc91 mc91 mc91 mc91 46
    mc91 mc91 mc91 mc91 mc91 mc91 57
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 68
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 79
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 90
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 101
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 91
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 102
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 92
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 103
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 93
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 104
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 94
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 105
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 95
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 106
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 96
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 107
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 97
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 108
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 98
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 109
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 99
    mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 mc91 110
```



```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91 mc91 mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 mc91 mc91 mc91 105
mc91 mc91 mc91 mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 mc91 mc91 mc91 106
mc91 mc91 mc91 mc91 mc91 mc91 mc91 96
mc91 mc91 mc91 mc91 mc91 mc91 mc91 107
mc91 mc91 mc91 mc91 mc91 mc91 mc91 97
mc91 mc91 mc91 mc91 mc91 mc91 mc91 108
mc91 mc91 mc91 mc91 mc91 mc91 mc91 98
mc91 mc91 mc91 mc91 mc91 mc91 mc91 109
mc91 mc91 mc91 mc91 mc91 mc91 mc91 99
mc91 mc91 mc91 mc91 mc91 mc91 mc91 110
mc91 mc91 mc91 mc91 mc91 mc91 mc91 100
mc91 mc91 mc91 mc91 mc91 mc91 mc91 111
mc91 mc91 mc91 mc91 mc91 mc91 mc91 101
mc91 mc91 mc91 mc91 mc91 mc91 mc91 91
mc91 mc91 mc91 mc91 mc91 mc91 mc91 102
mc91 mc91 mc91 mc91 mc91 mc91 mc91 92
mc91 mc91 mc91 mc91 mc91 mc91 mc91 103
mc91 mc91 mc91 mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 mc91 mc91 mc91 105
mc91 mc91 mc91 mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 mc91 mc91 mc91 106
mc91 mc91 mc91 mc91 mc91 mc91 mc91 96
mc91 mc91 mc91 mc91 mc91 mc91 mc91 107
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91 mc91 mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 mc91 mc91 mc91 106
mc91 mc91 mc91 mc91 mc91 mc91 mc91 96
mc91 mc91 mc91 mc91 mc91 mc91 mc91 107
mc91 mc91 mc91 mc91 mc91 mc91 mc91 97
mc91 mc91 mc91 mc91 mc91 mc91 mc91 108
mc91 mc91 mc91 mc91 mc91 mc91 mc91 98
mc91 mc91 mc91 mc91 mc91 mc91 mc91 109
mc91 mc91 mc91 mc91 mc91 mc91 mc91 99
mc91 mc91 mc91 mc91 mc91 mc91 mc91 110
mc91 mc91 mc91 mc91 mc91 mc91 mc91 100
mc91 mc91 mc91 mc91 mc91 mc91 mc91 111
mc91 mc91 mc91 mc91 mc91 mc91 mc91 101
mc91 mc91 mc91 mc91 mc91 mc91 mc91 91
mc91 mc91 mc91 mc91 mc91 mc91 mc91 102
mc91 mc91 mc91 mc91 mc91 mc91 mc91 92
mc91 mc91 mc91 mc91 mc91 mc91 mc91 103
mc91 mc91 mc91 mc91 mc91 mc91 mc91 93
mc91 mc91 mc91 mc91 mc91 mc91 mc91 104
mc91 mc91 mc91 mc91 mc91 mc91 mc91 94
mc91 mc91 mc91 mc91 mc91 mc91 mc91 105
mc91 mc91 mc91 mc91 mc91 mc91 mc91 95
mc91 mc91 mc91 mc91 mc91 mc91 mc91 106
mc91 mc91 mc91 mc91 mc91 mc91 mc91 96
mc91 mc91 mc91 mc91 mc91 mc91 mc91 107
mc91 mc91 mc91 mc91 mc91 mc91 mc91 97
mc91 mc91 mc91 mc91 mc91 mc91 mc91 108
mc91 mc91 mc91 mc91 mc91 mc91 mc91 98
mc91 mc91 mc91 mc91 mc91 mc91 mc91 109
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...  
mc91 mc91 mc91 mc91 97  
mc91 mc91 mc91 mc91 mc91 108  
mc91 mc91 mc91 mc91 98  
mc91 mc91 mc91 mc91 mc91 109  
mc91 mc91 mc91 mc91 99  
mc91 mc91 mc91 mc91 mc91 110  
mc91 mc91 mc91 mc91 100  
mc91 mc91 mc91 mc91 mc91 111  
mc91 mc91 mc91 mc91 101  
mc91 mc91 mc91 91  
mc91 mc91 mc91 mc91 102  
mc91 mc91 mc91 92  
mc91 mc91 mc91 mc91 103  
mc91 mc91 mc91 93  
mc91 mc91 mc91 mc91 104  
mc91 mc91 mc91 94  
mc91 mc91 mc91 mc91 105  
mc91 mc91 mc91 95  
mc91 mc91 mc91 mc91 106  
mc91 mc91 mc91 96  
mc91 mc91 mc91 mc91 107  
mc91 mc91 mc91 97  
mc91 mc91 mc91 mc91 108  
mc91 mc91 mc91 98  
mc91 mc91 mc91 mc91 109  
mc91 mc91 mc91 99  
mc91 mc91 mc91 mc91 110  
mc91 mc91 mc91 100  
mc91 mc91 mc91 mc91 111
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...  
mc91 mc91 mc91 mc91 110  
mc91 mc91 mc91 100  
mc91 mc91 mc91 mc91 111  
mc91 mc91 mc91 101  
mc91 mc91 91  
mc91 mc91 mc91 102  
mc91 mc91 92  
mc91 mc91 mc91 103  
mc91 mc91 93  
mc91 mc91 mc91 104  
mc91 mc91 94  
mc91 mc91 mc91 105  
mc91 mc91 95  
mc91 mc91 mc91 106  
mc91 mc91 96  
mc91 mc91 mc91 107  
mc91 mc91 97  
mc91 mc91 mc91 108  
mc91 mc91 98  
mc91 mc91 mc91 109  
mc91 mc91 99  
mc91 mc91 mc91 110  
mc91 mc91 100  
mc91 mc91 mc91 111  
mc91 mc91 101  
mc91 91  
mc91 mc91 102  
mc91 92  
mc91 mc91 103
```

```
file:///C:/Users/Gustavo Márquez/Documents/Visual Studio 2013/Projects/Funcion91JohnMcCarthy/Funcion91JohnMcCarthy/bin/Debug/Funcion91JohnMc...
mc91 mc91 mc91 110
mc91 mc91 100
mc91 mc91 mc91 111
mc91 mc91 101
mc91 91
mc91 mc91 102
mc91 92
mc91 mc91 103
mc91 93
mc91 mc91 104
mc91 94
mc91 mc91 105
mc91 95
mc91 mc91 106
mc91 96
mc91 mc91 107
mc91 97
mc91 mc91 108
mc91 98
mc91 mc91 109
mc91 99
mc91 mc91 110
mc91 100
mc91 mc91 111
mc91 101
91
McCarthy91 de 2 es 91
mc91( 3 ) =
    mc91 mc91 14
```